

Fu-Te Wong

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PROFILE

AI/ML engineer and researcher (10+ yrs) specialising in multi-agent self-improving LLM systems (alignment and interpretability), quantum-integrated ML, and deep RL. PhD candidate at University of Toronto (Vector Institute affiliate). Shipped end-to-end ML pipelines on cloud infra; experienced in cluster orchestration, MLOps, and infrastructure. Proven track record delivering AI research, building end-to-end ML pipelines.

TECHNICAL SKILLS

ML / AI	PyTorch, JAX, TensorFlow-Quantum, Scikit-learn, Hugging Face, LLM fine-tuning & agents, Multi-agent RL, LangChain, Qiskit, PennyLane
Stack	Python, C/C++, Java, SQL, JavaScript/TypeScript, React, Next.js, Django REST, Bash/Shell
CI/CD	Git, GitHub Actions, Docker, Docker Compose; experience with CI pipelines for ML model validation and automated test suites
Infra / MLOps	GCP, AWS (S3/EC2), Hadoop/Spark (CDH), TensorflowOnSpark, Linux sysadmin, cluster provisioning & job scheduling, vLLM, SGLang, megatron, verl
Data	NumPy, Pandas, SciPy; multimodal neuroimaging (fMRI/MRI/MEG/DTI); topological & dynamical systems analysis

EXPERIENCE

Researcher

2024 – Present

Vector Institute for Artificial Intelligence, Toronto

- Built multi-agent LLM/RL pipeline for automated QML algorithm search; abstracted a novel *advanced for-loop* construct accepted at QTML 2025 (arXiv:2506.18260)
- Designed agentic search framework with modular tool-use, prompt chaining, and self-evaluation loops; architect inference system; authored quantum backpropagation formulation (under review)
- Formulating artificial consciousness via Integrated Information Theory
- Co-organiser, AI-for-Reasoning reading group; driving LLM multi-token prediction on multi-hop reasoning and suicidality-detection agent projects; DPO variant (WIP)

Ph.D. Researcher — CS & Medical Science

2021 – 2026

University of Toronto (thesis: Agentic system for exploration of algorithmic ANNs)

- Contributed to generative quantum eigensolver (GQE) (arXiv:2401.09253)
- Engineered virtual brain stimulation AI; integrated E-field modelling with connectome tractography for TMS target selection

Teaching Assistant — Deep Learning

2020 – 2023

Neuromatch Academy (Lead TA 2021; global cohort of 1,000+ students)

- Delivered DL curriculum (PyTorch); led debugging sessions and model evaluation workshops across distributed online cohorts

Data Analyst

2019 – 2021

Institute of Linguistics, Academia Sinica, Taiwan

- Designed and deployed multimodal neuroimaging pipelines (MRI/fMRI/DTI/MEG) for cross-age working-memory study; reduced manual QC time by ~40% through automated topological analysis scripts

Research Assistant — Medical AI

2018-2019

Center for AI in Medicine, Taipei Medical University

- Delivered CNN-based diagnostic model for idiopathic normal-pressure hydrocephalus and brain-tumour segmentation; published CNN suicidality-prediction model (*J. Clinical Psychiatry*, 82(2), 2021)
- Built and documented end-to-end data pipeline: DICOM ingestion → preprocessing → model inference → clinical reporting

Software Engineer

2017

Cybersecurity Technology Institute, Institute for Information Industry, Taiwan

- Developed security tooling and data-processing scripts in Python/C++ within an agile team; participated in code review and issue tracking workflows

Visiting Student — Big Data & ML Infrastructure

2016 – 2017

Modeling & Informatics Lab, National Taiwan University

- Provisioned and administrated Hadoop/Spark CDH cluster for large-scale neural image analysis (Thunder-project, TensorflowOnSpark); aligned chatbot LU/NLG embeddings with brain activity signals

Clinical Psychologist

2011 – 2013

Taipei Veterans General Hospital

- Psychiatry Department, Neurological Institute, and Rehabilitation Center

SELECTED PROJECTS

MetaScientist (Meta_Scientist): Self-improving agentic research framework pairing LLM multi-agent orchestration with Monte-Carlo tree search (MCTS) to automatically search quantum machine-learning algorithm pipelines; integrates web search, task planning, and a dual-loop research cycle. Implementation framework behind the QTML 2025 paper; extends MetaGPT/SELA. *Tech*: Python, MetaGPT, MCTS, Docker

DDB Knowledge Agent (ddb-knowledge-agent-py): Streaming RAG agent fusing pgvector similarity retrieval with Postgres full-text search via reciprocal rank fusion (RRF); ReAct tool calling, query-complexity-tiered model routing, semantic caching, and cost auditing; PostgreSQL runs under rootless Apptainer, deployable on Docker-less HPC clusters. *Tech*: Python, FastAPI, SSE, Gemini, PostgreSQL/pgvector, Apptainer

GOAT Network Agent (goat-agent_demo): Autonomous Web3 agent registering an ERC-8004 on-chain identity and wired into the x402 payment flow; executes price tracking → conditional trigger → on-chain payment → attested settlement end-to-end with no human intervention after launch. *Tech*: Node.js, Express, ERC-8004, x402

optlab — Derivatives Pricing & Strategy Backtesting (optlab-py): Pricing library and vectorised backtester built to test whether the variance risk premium survives transaction costs. Black-Scholes analytic Greeks, CRR trees with American exercise,

variance-reduced Monte Carlo (12.7× via antithetic + control variates), Heston characteristic-function pricing and Andersen QE simulation; implied-vol inversion, raw-SVI and Heston surface calibration. Across 24 simulated 10-year paths: delta hedging lifts short-straddle Sharpe from 0.24 to 0.49, while frictions consume 70% of gross P&L institutionally and all of it at retail. 102 model-validation tests (convergence orders, Greeks vs finite differences, no-arbitrage bounds, P&L vs hand-computed ledger). *Tech: Python, NumPy, SciPy, Pandas, pytest, GitHub Actions*

EDUCATION

Ph.D. Candidate , Institute of Medical Science, University of Toronto, Canada	2021 – 2026
M.S. , Clinical Psychology, National Chung Cheng University, Taiwan	2009 – 2012

SELECTED PUBLICATIONS

Wong, F.-T. (2026). Quantum Backpropagation. (*Under review*).

Frydenlund, A., Wong, F.-T., et al. (2026). Multi-Token Prediction and Alternative Tokenization Improve Multi-Hop Reasoning and Graph Search. (*Under review*).

Wong, F.-T. (2025). Advanced For-Loop for QML algorithm search. *arXiv:2506.18260*. QTML 2025.

Nakaji, K., ..., Wong, F.-T., et al. (2024). GQE. *arXiv:2401.09253*.

Chen, V.C.-H., Wong, F.-T., et al. (2021). CNN suicidality prediction. *J. Clin. Psychiatry* 82(2).

AWARDS

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- Vector Institute Research Scholarship (2024, 2025)
 - Pixnet Hackathon: A.I. Image Gen. (2018) - 2nd Prize
 - Pixnet Hackathon: A.I. Cloze (2016) - 1st Prize